

Roll No.

TMA-303

**B. TECH. (MECHANICAL ENGINEERING)
(THIRD SEMESTER)
MID SEMESTER EXAMINATION, Oct., 2022
ENGINEERING MATHEMATICS—III**

Time : 1½ Hours

Maximum Marks : 50

Note : (i) Answer all the questions by choosing any *one* of the sub-questions.
(ii) Each sub-question carries 10 marks.

1. (a) Show that the function $u(x, y) = x^4 - 6x^2y^2 + y^4$ is harmonic. Also, find its harmonic conjugate function. (CO1)

OR

- (b) Using Cauchy's integral formula to compute the value of the integral $\oint_C \frac{\sin \pi z^2 + \cos \pi z^2}{(z-1)(z-2)} dz$, where $C : |z| = 3$. (CO2)

2. (a) Show that the function $f(z) = |xy|^{\frac{1}{2}}$ is not analytic function at the origin although the C-R equations are satisfied at that point. (CO1)

OR

- (b) Determine the bilinear transformation which maps the points 2, i , -2 into the points 1, i , -1 . (CO2)

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(2)

3. (a) Using Milne-Thomson method to construct the analytic function $f(z) = u(x, y) + iv(x, y)$, where the real part of $f(z)$ is $u(x, y) = e^{-x}[(x^2 - y^2) \cos y + 2xy \sin y]$. (CO1)

OR

- (b) Check the continuity and differentiability of the following function at $z = 0$: (CO1)

(i) $f(z) = \bar{z}$

(ii) $f(z) = z\bar{z}$

4. (a) Show that the function $f(z) = |z|^2$ is not analytic at $z = 0$. (CO1)

OR

- (b) Using Cauchy's integral formula to evaluate the integral

$$\oint_C \frac{z^2 - z - 1}{z(z-1)^2} dz, \text{ where } C : \left| z - \frac{1}{2} \right| = 1. \quad (\text{CO2})$$

5. (a) Evaluate the integral $\int_0^{1+i} (x - y + ix^2) dz$ along: (CO2)

(i) the straight line from $z = 0$ to $z = 1 + i$.

(ii) the real axis from $z = 0$ to $z = 1$ and then along a line parallel to imaginary axis from $z = 1$ to $z = 1 + i$.

OR

- (b) Show that the function $f(z) = z + 2\bar{z}$ is not analytic anywhere in the complex plane. (CO1)